SHRIHARI M V

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Professional Summary

Fresher Software Engineer with hands-on experience building full-stack, backend, and AI-powered applications using Python, React, FastAPI, REST APIs, and SQL. Built independently developed systems spanning authenticated web workflows, APIs, persistent data, stateful workflows, retrieval pipelines, testing, validation, and telemetry. Strong foundation in software engineering and debugging, with experience taking systems from design and implementation through testing and iterative improvement.

Technical Skills

Languages	Python, Java, SQL
Core Engineering	Data Structures & Algorithms, Object-Oriented Programming, API Design, REST APIs, Async Programming, Git, GitHub
Backend	FastAPI, Pydantic, Authentication, Request/Response Validation, Swagger/OpenAPI
Frontend	React, React Router, TanStack Query
Databases	SQL, MongoDB, Data Modeling, CRUD, Querying
Testing & Reliability	Pytest, Integration Testing, Error Handling, Structured Logging, Telemetry
AI Systems	LangGraph, LangChain, RAG, Tool Calling, Embeddings, Semantic Search
Deployment	Docker, MLflow, ONNX Runtime

Projects

AttriShield – Employee Attrition Decision-Support Platform GitHub

React • TypeScript • FastAPI • Python • XGBoost

- Built an end-to-end employee attrition decision-support application connecting a **React/TypeScript frontend** with FastAPI services for prediction, batch analysis, and analytics workflows.
- Implemented **JWT/Bearer authentication, protected routes, controlled forms, centralized API communication, and response validation** to connect frontend workflows with authenticated backend services.
- Integrated **XGBoost inference and SHAP explanations** into application workflows, exposing prediction results through interactive product interfaces rather than a standalone ML notebook.
- Added **server-state management, asynchronous loading/error handling, MLflow tracking, and application telemetry** to make model-driven workflows testable and observable.

StudyMentor – Adaptive Learning Platform GitHub

Python • FastAPI • MongoDB • Pydantic • LangGraph

- Built an adaptive learning platform with a **FastAPI backend, asynchronous services, MongoDB persistence, and deterministic learning workflows** for practice, review scheduling, analytics, and learner memory.
- Separated **API routing, validation, business logic, persistence, and agent orchestration** so core application behavior remained independent of LLM-driven workflows.
- Engineered controlled agent execution using **typed state, tool calling, conditional routing, bounded retries, checkpointing, persistent memory, and human-review escalation** for failure-aware workflow handling.
- Validated backend and agent behavior with **19 passing pytest tests** and integration experiments covering tool calling, learner-memory loading, adaptive planning, and workflow execution.

CarnaticPrep – Retrieval-Augmented Knowledge System GitHub

Python • FastAPI • ChromaDB • Sentence Transformers • CrossEncoder

- Designed a hybrid retrieval pipeline combining **dense retrieval, BM25, adaptive query routing, weighted Reciprocal Rank Fusion (RRF), and CrossEncoder reranking** for domain-specific knowledge search.
- Built corpus-aware indexing workflows using **structure-aware chunking, metadata normalization, and adaptive PDF/CSV processing** across heterogeneous knowledge sources.
- Developed configurable **FastAPI retrieval and generation services** supporting vector indexing, retrieval workflows, grounded response generation, and Swagger/OpenAPI documentation.
- Added **structured logging** across ingestion and retrieval workflows to make query behavior and system execution easier to inspect and debug.

Education

B.E. Computer Science & Engineering (Data Science) 2022 – 2026
Bangalore Institute of Technology CGPA: **8.6/10**

PCMB 2020 – 2022
Sri Kumarans PU College Percentage: **94.83%**

Experience

MindMatrixEd – Generative AI Intern Feb 2026 – May 2026

- Explored AI-assisted Android application workflows using Google AI Studio, Android Studio and Kotlin contributing to feature prototyping and user-centric product design.